

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

1. Scenario Understanding

A newly established Smart Campus has been assigned the IPv4 network 192.168.100.0/24 to design a secure and efficient network infrastructure for different academic departments. The objective is to allocate separate subnets for the School of Computing, School of Electronics, and School of Mechanical Engineering using the given subnet masks. The subnet design should determine the network address, broadcast address, valid host IP address range, usable host addresses, and identify the remaining unused IP address range for future expansion.

2. Given Data

Original Network : 192.168.100.0/24

Allocated Subnets

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

3. Identification of Key Issues

- Divide the given /24 network into three subnets.
- Allocate subnets sequentially without overlap.
- Determine the network address of each subnet.
- Determine the broadcast address.
- Identify the valid host IP address range.
- Calculate the usable host addresses.
- Determine the remaining unused address range.

4. Formula Used

Host Bits

Host Bits = 32 – Prefix Length

Total Addresses

Total Addresses = $2^{(\text{Host Bits})}$

Usable Hosts

Usable Hosts = $2^{(\text{Host Bits})} - 2$

Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$

5. Step-by-Step Solution

A) School of Computing

Prefix Length

/25

Subnet Mask

255.255.255.128

Host Bits

$32 - 25 = 7$

Block Size

$256 - 128 = 128$

Total Addresses

$2^7 = 128$

Usable Hosts

$128 - 2 = 126$

Network Address

192.168.100.0

Valid Host Range

192.168.100.1 – 192.168.100.126

Broadcast Address

192.168.100.127

B) School of Electronics

Prefix Length

/26

Subnet Mask

255.255.255.192

Host Bits

$32 - 26 = 6$

Block Size

$256 - 192 = 64$

Total Addresses

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Next Available Network

192.168.100.128

Network Address

192.168.100.128

Valid Host Range

192.168.100.129 – 192.168.100.190

Broadcast Address

192.168.100.191

C) School of Mechanical Engineering**Prefix Length**

/27

Subnet Mask

255.255.255.224

Host Bits

$$32 - 27 = 5$$

Block Size

$$256 - 224 = 32$$

Total Addresses

$$2^5 = 32$$

Usable Hosts

$$32 - 2 = 30$$

Next Available Network

192.168.100.192

Network Address

192.168.100.192

Valid Host Range

192.168.100.193 – 192.168.100.222

Broadcast Address

192.168.100.223

6. Remaining Unused IP Address Range

Starting Address

192.168.100.224

Ending Address

192.168.100.255

This address space remains available for future departments or network expansion.

7. Decision Making

The subnets have been allocated from the largest subnet to the smallest subnet to maximize IP address utilization and avoid address fragmentation. Each department receives an independent subnet, reducing broadcast traffic, improving security, and simplifying network management. The remaining address space is reserved for future scalability.

8. Final Result

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	255.255.255.128 (/25)	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	255.255.255.192 (/26)	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	255.255.255.224 (/27)	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Remaining Unused Address Range

9. Engineering Interpretation

The Smart Campus network has been successfully divided into three independent subnets according to the specified requirements. The School of Computing receives 126 usable host addresses, the School of Electronics receives 62 usable

host addresses, and the School of Mechanical Engineering receives 30 usable host addresses. The remaining address range from 192.168.100.224 to 192.168.100.255 is reserved for future expansion. This subnetting strategy improves network performance, reduces broadcast traffic, enhances security through network segmentation, and provides an efficient and scalable network design.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

1. Scenario Understanding

A multinational software company has been allocated the private IPv4 network 10.10.0.0/16 for its headquarters. Different departments require different subnet sizes based on the number of employees. The objective is to allocate suitable subnets for Software Development, Quality Assurance, Human Resources, and Executive Management by determining the network address, broadcast address, valid host IP address range, usable host addresses, and the remaining unused IP address range.

2. Given Data

Original Network : 10.10.0.0/16

Allocated Subnets

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

3. Identification of Key Issues

- Divide the given /16 network into four departmental subnets.
- Allocate subnets sequentially without overlap.
- Determine the network and broadcast address of each subnet.
- Identify the valid host IP address range.
- Calculate the usable host addresses.
- Determine the remaining unused address range.

4. Formula Used

Host Bits

Host Bits = $32 - \text{Prefix Length}$

Total Addresses

Total Addresses = $2^{(\text{Host Bits})}$

Usable Hosts

Usable Hosts = $2^{(\text{Host Bits})} - 2$

Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$

5. Step-by-Step Solution

A) Software Development

Prefix Length

/24

Subnet Mask

255.255.255.0

Host Bits

$32 - 24 = 8$

Block Size

$$256 - 0 = 256$$

Total Addresses

$$2^8 = 256$$

Usable Hosts

$$256 - 2 = 254$$

Network Address**10.10.0.0****Valid Host Range****10.10.0.1 – 10.10.0.254****Broadcast Address****10.10.0.255****B) Quality Assurance****Prefix Length****/25****Subnet Mask****255.255.255.128****Host Bits**

$$32 - 25 = 7$$

Block Size

$$256 - 128 = 128$$

Total Addresses

$$2^7 = 128$$

Usable Hosts

$$128 - 2 = 126$$

Next Available Network**10.10.1.0****Network Address**

10.10.1.0

Valid Host Range

10.10.1.1 – 10.10.1.126

Broadcast Address

10.10.1.127

C) Human Resources

Prefix Length

/26

Subnet Mask

255.255.255.192

Host Bits

$32 - 26 = 6$

Block Size

$256 - 192 = 64$

Total Addresses

$2^6 = 64$

Usable Hosts

$64 - 2 = 62$

Next Available Network

10.10.1.128

Network Address

10.10.1.128

Valid Host Range

10.10.1.129 – 10.10.1.190

Broadcast Address

10.10.1.191

D) Executive Management

Prefix Length

/27

Subnet Mask

255.255.255.224

Host Bits

$32 - 27 = 5$

Block Size

$256 - 224 = 32$

Total Addresses

$2^5 = 32$

Usable Hosts

$32 - 2 = 30$

Next Available Network

10.10.1.192

Network Address

10.10.1.192

Valid Host Range

10.10.1.193 – 10.10.1.222

Broadcast Address

10.10.1.223

6. Remaining Unused IP Address Range

Starting Address

10.10.1.224

Ending Address

10.10.255.255

The remaining address space can be reserved for future departments, branch offices, or network expansion.

7. Decision Making

The subnet allocation follows the largest-to-smallest approach, ensuring efficient utilization of the available address space. Each department receives an independent subnet based on its requirements, reducing broadcast traffic, improving security, and supporting future scalability.

8. Final Result

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	255.255.255.0 (/24)	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	255.255.255.128 (/25)	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	255.255.255.192 (/26)	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	255.255.255.224 (/27)	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Remaining Unused Address Range

10.10.1.224 – 10.10.255.255

9. Engineering Interpretation

The headquarters network has been successfully divided into four independent subnets according to the departmental requirements. Software Development supports 254 usable hosts, Quality Assurance supports 126 usable hosts, Human Resources supports 62 usable hosts, and Executive Management supports 30 usable hosts. The remaining address range from 10.10.1.224 to 10.10.255.255 is reserved for future expansion. This subnet allocation improves network performance, minimizes broadcast traffic, enhances security, and provides an efficient and scalable network infrastructure.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.

2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

1. Scenario Understanding

A smart hospital is upgrading its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices, healthcare systems, and IoT sensors. The assigned IPv6 network is 2001:0db8:abcd:0000:0000:0000:0000/64. The objective is to compress the IPv6 address, expand it back to its original representation, identify the network prefix, and calculate the total number of host addresses supported by the subnet.

2. Given Data

IPv6 Network

2001:0db8:abcd:0000:0000:0000:0000/64

Prefix Length

/64

3. Identification of Key Issues

- Convert the IPv6 address into compressed notation.
- Expand the compressed address into full hexadecimal notation.
- Identify the network prefix using the /64 prefix.
- Calculate the number of host bits.
- Determine the total number of possible host addresses.

4. Formula Used

IPv6 Compression Rules

- Remove leading zeros from each hexadecimal block.
- Replace one continuous sequence of consecutive zero blocks with "::".
- "::" can appear only once in an IPv6 address.

Host Bits

Host Bits = 128 – Prefix Length

Total Host Addresses

Total Host Addresses = $2^{(\text{Host Bits})}$

5. Step-by-Step Solution

Step 1

Original IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Step 2

Remove Leading Zeros

2001:db8:abcd:0:0:0:0:0

Step 3

Compress Consecutive Zero Blocks

Compressed IPv6 Address

2001:db8:abcd::

Step 4

Expand the Compressed Address

Expanded IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Step 5

Identify the Network Prefix

Prefix Length

/64

Network Prefix

2001:0db8:abcd:0000::/64

Step 6

Calculate Host Bits

Host Bits

$128 - 64$

$= 64$

Step 7

Calculate Total Host Addresses

Total Host Addresses

2^{64}

= 18,446,744,073,709,551,616

Approximately

18.4 Quintillion Host Addresses

6. Decision Making

A /64 prefix is the standard subnet size in IPv6 networking because it supports Stateless Address Auto Configuration (SLAAC), efficient routing, and compatibility with modern networking devices. IPv6 provides an enormous address space, making it ideal for smart hospitals with thousands of interconnected medical devices and IoT sensors.

7. Final Result

Parameter / Field	Specification / Value
Original IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Compressed IPv6 Address	2001:db8:abcd::
Expanded IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Network Prefix	2001:0db8:abcd:0000::/64
Prefix Length	/64
Host Bits	64
Total Possible Host Addresses	18,446,744,073,709,551,616 (2^{64})

8. Engineering Interpretation

The hospital network has been successfully migrated to IPv6 addressing using the assigned /64 network prefix. The IPv6 address has been correctly compressed and expanded according to IPv6 addressing rules. With 64 host bits available, the subnet supports 18,446,744,073,709,551,616 unique host addresses, providing sufficient scalability for future medical devices, IoT sensors, cloud-based healthcare systems, and advanced smart hospital infrastructure.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

1. Scenario Understanding

An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains routing table entries for different destination networks. A packet arrives with the destination IP address 192.168.2.45. The objective is to determine the destination subnet, identify the matching routing table entry, determine the next-hop network/interface, and specify the default route if no routing table entry matches.

2. Given Data

Routing Table Entries

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP Address

192.168.2.45

3. Identification of Key Issues

- Identify the destination subnet of the given IP address.

- Find the matching routing table entry.
- Determine the interface used to forward the packet.
- Identify the default route used when no matching entry exists.

4. Formula Used

Network Address

Network Address = IP Address AND Subnet Mask

Subnet Mask for /24

255.255.255.0

Routing Principle

A router forwards packets using the Longest Prefix Match (LPM). The routing table entry with the most specific matching prefix is selected for forwarding.

5. Step-by-Step Solution

Step 1

Destination IP Address

192.168.2.45

Step 2

Subnet Mask

255.255.255.0 (/24)

Step 3

Determine the Destination Network

Applying the subnet mask:

192.168.2.45

AND

255.255.255.0

Result

192.168.2.0

Therefore, the destination subnet is

192.168.2.0/24

Step 4

Matching Routing Table Entry

The routing table contains the following entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

The destination network matches

192.168.2.0/24

Step 5

Next-Hop Network / Interface

Since the destination network is directly available in the routing table, the packet is forwarded through the interface connected to the

192.168.2.0/24

network.

Step 6

Default Route

If the destination network does not match any routing table entry, the router forwards the packet using the default route.

Default Route

0.0.0.0/0

6. Decision Making

The router correctly identifies the destination network using the subnet mask and selects the matching routing table entry based on the Longest Prefix Match principle. If no matching route exists, the default route ensures that packets are forwarded toward another router instead of being discarded immediately.

7. Final Result

Parameter / Field	Details / Value
Destination IP Address	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Routing Table Entry	192.168.2.0/24

Next-Hop Network / Interface	Interface connected to the 192.168.2.0/24 network
Default Route	0.0.0.0/0

0.0.0.0/0

8. Engineering Interpretation

The router successfully identifies that the destination IP address 192.168.2.45 belongs to the subnet 192.168.2.0/24. Based on the routing table, the packet is forwarded through the interface connected to this network. If a destination network is unavailable in the routing table, the router uses the default route (0.0.0.0/0) to forward the packet to another router. This routing mechanism improves communication efficiency, supports scalable network connectivity, and ensures reliable packet delivery across interconnected networks.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

1. Scenario Understanding

A company has established two office departments, Administration and Finance, connected through a Layer-3 router. Each department is assigned a separate

subnet with a dedicated gateway. The objective is to determine the network address, broadcast address, valid host IP address range, suggest suitable IP addresses for three computers in each LAN, and identify the default gateway used for communication.

2. Given Data

Administration LAN

Network Address : 192.168.10.0/26

Gateway : 192.168.10.1

Finance LAN

Network Address : 192.168.10.64/26

Gateway : 192.168.10.65

3. Identification of Key Issues

- Determine the network address of each LAN.
- Determine the broadcast address.
- Identify the valid host IP address range.
- Assign suitable IP addresses for three computers in each LAN.
- Identify the correct default gateway for each department.

4. Formula Used

Host Bits

Host Bits = $32 - \text{Prefix Length}$

Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$

Total Addresses

Total Addresses = $2^{(\text{Host Bits})}$

Usable Hosts

Usable Hosts = $2^{(\text{Host Bits})} - 2$

5. Step-by-Step Solution

A) Administration LAN

Prefix Length

/26

Subnet Mask

255.255.255.192

Host Bits

$$32 - 26 = 6$$

Block Size

$$256 - 192 = 64$$

Total Addresses

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Network Address

192.168.10.0

Broadcast Address

192.168.10.63

Valid Host Range

192.168.10.1 – 192.168.10.62

Suggested Computer IP Addresses

Computer 1 : 192.168.10.2

Computer 2 : 192.168.10.3

Computer 3 : 192.168.10.4

Default Gateway

192.168.10.1

B) Finance LAN**Prefix Length**

/26

Subnet Mask

255.255.255.192

Host Bits

$$32 - 26 = 6$$

Block Size

$$256 - 192 = 64$$

Total Addresses

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 – 192.168.10.126

Suggested Computer IP Addresses

Computer 1 : 192.168.10.66

Computer 2 : 192.168.10.67

Computer 3 : 192.168.10.68

Default Gateway

192.168.10.65

6. Decision Making

The two departments are configured in separate /26 subnets, allowing efficient communication through a Layer-3 router. Separate subnets reduce broadcast traffic, improve network security, simplify management, and enable controlled communication between departments using their respective gateway interfaces.

7. Final Result

Department / LAN	Network Address	Broadcast Address	Valid Host Range	Default Gateway	Suggested Computer IP Addresses
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62	192.168.10.1	192.168.10.2, 192.168.10.3, 192.168.10.4
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126	192.168.10.65	192.168.10.66, 192.168.10.67, 192.168.10.68

8. Engineering Interpretation

The company network has been successfully divided into two independent LANs using /26 subnetting. Administration and Finance each receive 62 usable host addresses and communicate through a Layer-3 router using their respective default gateways. This network design reduces broadcast traffic, improves security through subnet isolation, simplifies administration, and provides a scalable infrastructure for future organizational growth.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

